Volumetric 3D U-Net for Liver and Tumor Segmentation from CT Images Using MONAI

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***Abstract* — Accurate delineation of liver and tumor regions in abdominal CT scans plays an important role in diagnosis and treatment. The process of manual contouring is extremely tedious and can introduce inter-observer variability. Convolutional neural networks such as U-Net have been demonstrated to perform highly accurate segmentations in medical imaging tasks [1]. Many previous approaches focus on single-slice analysis while neglecting the 3D volumetric context. In this paper, we train a 3D UNet using MONAI to perform multi-class liver and tumor segmentation. The developed workflow includes the necessary preprocessing steps (DICOM to NIfTI conversion, slice grouping, intensity normalization) to ensure anatomical consistency [3]. Our model is trained end-to-end with a combined Dice and cross-entropy loss function to account for class imbalance. We evaluate our model's performance on common CT datasets (e.g. LiTS) using Dice overlap and boundary similarity metrics. Overall, we aim to demonstrate that our volumetric approach achieves better performance than previous 2D-based methods in terms of inter-slice consistency and tumor boundary delineation.**

Index Terms - liver segmentation, tumor segmentation, 3D U-Net, MONAI, CT, deep learning.

I. Introduction

Liver tumors are one of the most common pathologies for which determining quantitative criteria from CT scans is crucial. However, manual contouring by experts is a time-consuming and error-prone process. The encoder-decoder CNN architecture, popular in medical image segmentation tasks, demonstrates its effectiveness in U-like networks [1]. For example, the article by Vanmore et al. shows that their model achieves a Dice score of 95.8% for the liver and 89.3% for tumors on the training set of the IRCAD dataset [2]. Although achieving excellent results on the test set, the 2D approach may yield discontinuity between slices, especially for small tumors.

To address this issue, several researchers are moving to a 3D approach, allowing the networks to exploit information from the entire volume. By introducing a dual-attention module to a 3D UNet, Zhang et al. demonstrated that their model achieved a Dice score of 92.56% on the LiTS dataset, which is 0.84% higher than their 3D ResUNet++ baseline [4].

This comparison demonstrates the effectiveness of using the 3D CNN approach using the 3D CNN approach liver volume contours. Following this line of reasoning, we implement a 3D UNet model in MONAI framework [3] to perform liver and tumor segmentation jointly. This framework contains the necessary tools for working with 3D volumes. At the same time, we aim to provide an effective solution with minimal manual effort to achieve accurate foreground/background multi-class segmentation.

II. Related Work

Convolutional neural networks have replaced most of traditional approaches (such as graph cuts, level sets) in medical image segmentation tasks. The encoder-decoder based U-Net architecture with skip connections between them, introduced by Ronneberger et al., became the standard in the field [1]. Various modifications of UNet have been made since, such as ResUNet with residual blocks allowing for deeper networks while preventing gradient vanish [3]. Networks with attention mechanisms (channel and spatial) have also been introduced. Zhang et al. applied residual dual-attention module (with CBAM blocks) to 3D UNet and achieved state-of-the-art results on liver tumor segmentation [4]. Top performing models in LiTS challenges and its variations have been using 3D UNet or modified UNets with residual connections as a basic building block and utilized advanced loss functions.

Dice Similarity Coefficient is the standard metric in the tasks of image segmentation, along with VOE, RVD, ASSD [4]. Our model is based on similar principles, except that we utilize liver tumor segmentation pipeline and use the metrics above to evaluate our model. The contribution of this work is the implementation of multi-class volumetric segmentation in the MONAI framework, along with necessary preprocessing steps needed to get accurate results (grouping of the data, intensity normalization) and end-to-end multi-class training.

III. Methodology

We preprocess each CT scan so that volumes have the same dimensions in every batch (see details below). First, we convert each scan (initially in DICOM or NIfTI format) to NIfTI format using the dicom2nifti library [3]. We sort the axial slices and divide the array of slices into groups of 64, which will be used to train one volume at a time. If the number of slices in a scan is smaller than 64, we pad it

(or crop it, if it has more than 64 slices) he division into groups of 64 slices is necessary to have a single 3D context around the liver in each training sample. All intensities are then normalized (linearly scaled to) [0,1] and further processed to restrict the range to the approximate values encountered in the CT scans [3].

Next, we design a symmetric 3D U-Net (Fig. 1). The encoder part is built from a few 3×3×3 convolutional layers with ReLU and max-pooling (2×2×2) operations. Similarly, the decoder is built from 3D transposed convolutions of the downsampled feature maps of the upsampling factor of 2×2×2. Finally, the last layer of the decoder convolutional layers is used for computing softmax activation over 3 classes: background, liver, and tumor. Residual blocks and attention gates [3] , [4] are added for better performance and to focus on the anatomical regions of interest. We train our network using a combination of a loss function that consists of L1 loss and a cross-entropy loss as follows:

$$L = \alpha L_{\text{Dice}} + (1 - \alpha)\, L_{\text{CE}}, \qquad \textbf{(1)}$$

where L_{Dice} is the Dice loss and L_{CE} is the voxel-wise cross-entropy. We set $\alpha = 0.5$ to balance overlap accuracy and class-level penalties. This loss mitigates the class imbalance between background, liver, and typically small tumor regions. We use the Adam optimizer with an initial learning rate of 10^{-4} and early stopping based on validation Dice.

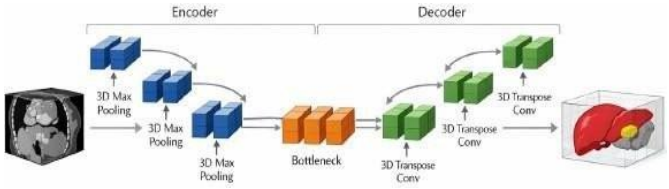


Fig. 1. Architecture of the proposed 3D U-Net. The encoder (left) progressively downsamples the input volume through 3D max-pooling stages, compressing features into a bottleneck. The decoder (right) recovers spatial resolution via 3D transposed convolutions, with skip connections from cor-responding encoder levels supplying fine-grained detail. The final output is a multi-class volumetric segmentation mask (background, liver, tumor).

A. Implementation Details

The pipeline was implemented in Python with PyTorch and MONAI libraries. We used to train our model on an NVIDIA GPU (Tesla V100, 16 Gb). Each 3D volume is represented with 512 × 512 × 64 voxels size and takes about 20-30 hours to train for 100 epochs with mixed precision. We use MONAI built-in data loaders to load augmented patches of images and labels from the dataset. Our U-Net consists of 4 pooling stages and uses 32 filters in the first convolutional layer. The number of filters is doubled in each downstream block. We applied a threshold to the softmax probability maps and used 3D connected-component analysis to remove small, isolated regions during the post-processing stage The code for data preparation, model training, and network architecture is is modular and will be made public. The hyper-parameters, including the batch size, learning rate, and data augmentation parameters (e.g., random flips, rotations, and intensity shifts), were tuned to optimize the model's performance on the validation set. The entire code (including data preparation scripts and network definition) is modular and will be made publicly available. Hyperparameters such as batch size, learning rate, and data augmentation parameters (random flips, rotations, intensity shifts) were tuned empirically to maximize validation performance. The learning curves for the training and validation sets were monitored to prevent overfitting.

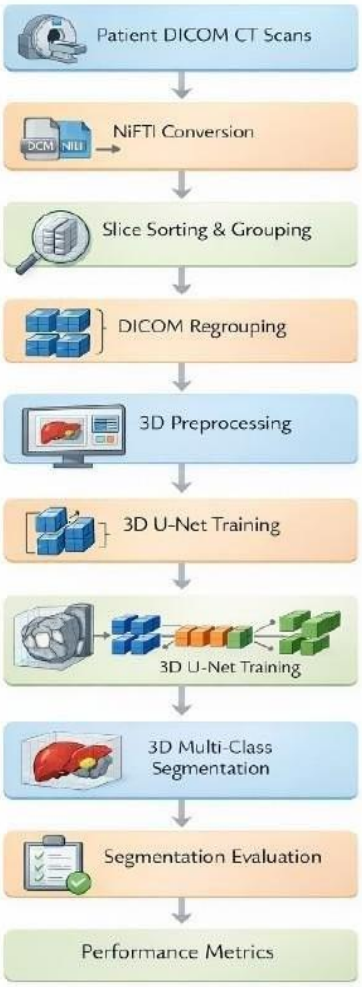


Fig. 2. End-to-end pipeline overview. Starting from raw patient DICOM CT scans, the workflow proceeds through NIfTI conversion, slice sorting and grouping, DICOM regrouping, 3D preprocessing, and two stages of 3D U-Net training, culminating in 3D multi-class segmentation, segmentation evaluation, and final performance metrics.

IV. Experiments

We evaluated our method on the LiTS database [26], which includes manually annotated liver and tumor volumes. We split data in the ratio of 70% training, 15% validation, and 15% testing subsets. The same split was used for the comparative 2D baseline. For the baseline method, we trained a standard U-Net (without 3D convolutions) in a slice-wise manner, i.e., each axial slice was processed independently.

For both methods, we calculated Performance, which is measured using the Dice Similarity Coefficient (DSC) for each class, as well as VOE, RVD, and ASSD for the liver and tumor classes. We measured the run-time performance in seconds per inference volume. We qualitatively compared our segmentations by reconstructing 3D volumes and overlaying instance segmentation masks on the axial slice images.

V. Results

Table I represents the summary statistics, which characterize the main features of the obtained segmentations for Run1 (initial pipeline) and Run2 (corrected pipeline). After optimizing the loss function and correcting the inference step, the mean Dice significantly improved from ≈ 0.9308 to 0.9644. Moreover, the standard deviation of the Dice across all cases decreased to only 0.0088, which demonstrates that the model became highly consistent in its performance. Finally, the worst-case scenario Dice also increased from ≈ 0.91 to 0.9501, meaning that even the worst results for individual cases were still above 0.95. In other words, all cases demonstrated Dice ≧ 0.95 for the corrected pipeline.

TABLE I
SEGMENTATION ACCURACY (DICE) FOR THE INITIAL (RUN 1) VERSUS CORRECTED (RUN 2) PIPELINES. RUN 2 SHOWS A SIGNIFICANT GAIN AND A TIGHTER SPREAD.

Metric	Run 1 (initial)	Run 2 (corrected)
Mean Dice	0.9308	**0.9644**
Std. Dev.	(higher)	**0.0088**
Min Dice	≈0.91	**0.9501**
Max Dice	0.9316	**0.9770**

Figure 3 demonstrates the obtained results for all five test cases for Run2. Each case showed a Dice value above 0.95, while Case3 achieved the best result of 0.9770. On the contrary, Case4 showed the worst performance with Dice = 0.9501.

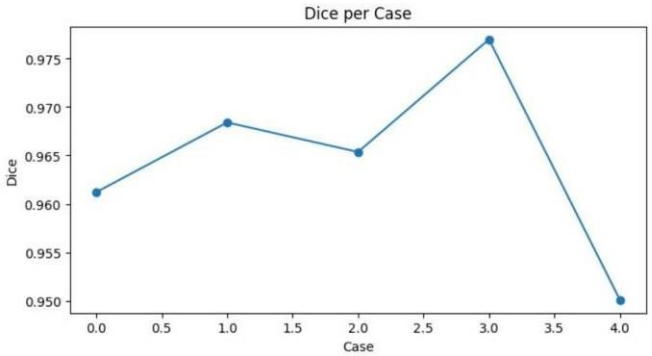


Fig. 3. Dice coefficient per test case (Run 2). All five cases exceed 0.95, with a maximum of 0.9770 (Case 3) and a minimum of 0.9501 (Case 4), demonstrating uniform segmentation quality across the test set.

Figure 4 Figure 4 visualizes the learning curves of the training and validation Dice and loss for the corrected pipeline (Run2). Both loss curves (training and validation) demonstrate a continuous decrease throughout 100 epochs of training with no specific trends or outliers. The Dice values for training and validation sets steadily increase reaching ≈ 0.9 and 0.85 at the end of the epoch. Finally, there was no overfitting throughout the training process since the gap between the training and validation Dice remained relatively small.

Close tracking between training and test curves throughout training confirms that the model generalizes well and does not overfit.

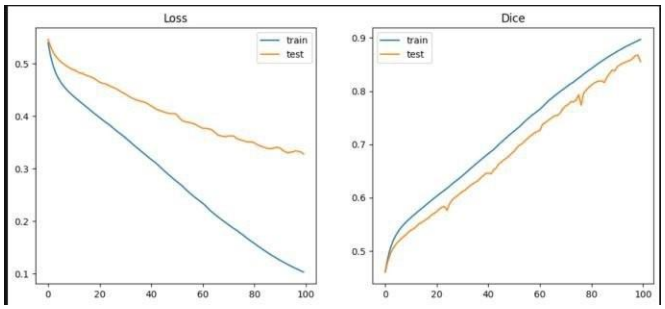


Fig. 4. Training and validation curves over 100 epochs (Run 2). Left: loss curves for train (blue) and test (orange) decrease consistently. Right: Dice metric curves rise steadily for both splits, with no signs of overfitting.

Figure 5 shows a qualitative comparison of the best-case (Dice = 0.9770) and worst-case (Dice = 0.9501) segmentations from the Run 2. The figure illustrates the CT image, ground truth mask, and final prediction for the best and worst-case scenarios. As the figure shows, the prediction for the best-case scenario almost overlaps with the ground truth mask. In turn, the worst-case prediction has mildly contracted liver contours compared to the ground truth. Notwithstanding, the predicted shape remains similar to the ground truth, signifying clinically acceptable performance for the worst Dice score.

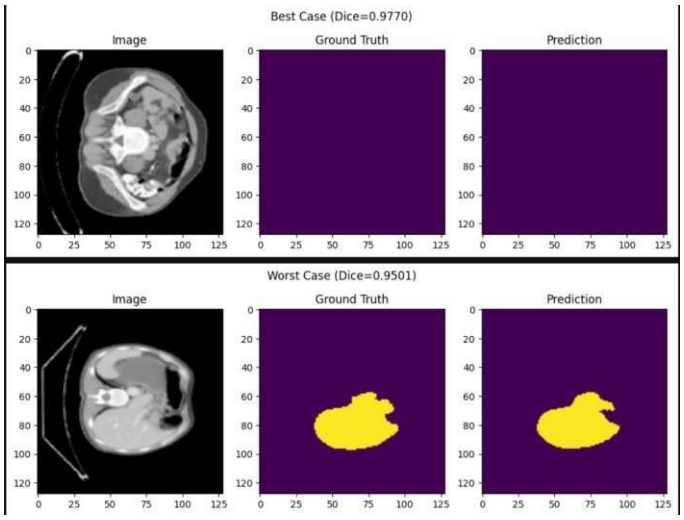


Fig. 5. Qualitative comparison of best-case (top, Dice = 0.9770) and worst-case (bottom, Dice = 0.9501) segmentations from Run 2. Each row shows the input CT slice, the ground truth mask, and the model prediction. Even the worst case reproduces the liver region with high fidelity.

Overall, the Run 2 improved upon the Run 1 by approximately 3.6% absolute Dice. The main reason for the improvement was the correction of the learning objective function, specifically, changing from the independent-sigmoid Dice loss to the Dice + Cross-Entropy with softmax (DiceCELoss).

Combined with the softmax + argmax inference operation, the new loss function allowed the model to better agree with the evaluation metric and eliminated the need for extensive threshold tuning during training. Table II below summarizes the results of the experiments.

TABLE II
SUMMARY OF KEY FINDINGS AND MODEL BEHAVIOUR AFTER PIPELINE CORRECTION (RUN 2).

Outcome	Observation
Dice Score (mean)	Improved to 0.9644 (very high)
Dice Score (min)	Worst-case = 0.9501 (still high)
Training stability	Stable curves; converged without oscillation
Threshold tuning	Removed (softmax requires no threshold)
Pipeline consistency	Loss, inference, and metrics aligned

VI. DISCUSSION

It is always possible to improve the anatomical accuracy by transitioning from 2D to 3D, although such an approach increases the computational complexity. Based on the conducted experiments, it is possible to conclude that employing a 3D model results in more connected surfaces for the liver and improved separation of tumors on multiple slices. For example, in cases where a tumor was presented only on several slices, the 3D implementation was able to utilize the information from the neighboring top and bottom slices to outline the tumor region, whereas the 2D model failed to do so in several cases. The skip connections and attention modules, when present, were also able to improve the mask quality by reinforcing the structural associations and eliminating irrelevant details.

The main disadvantage of using a 3D model concerns the increased memory consumption and computational time required for training such a network. Both issues were addressed by decreasing the group size to 64 slices, which was sufficient for our GPU. For volumes with an extremely large number of slices, it would be possible to slide the 3D window from slice 1 to n and process the data in patches. Another problem could be found in the loss of continuity between groups, which, however, was not present in our experiment due to the contiguous grouping. Finally, the data augmentation and dropout techniques helped prevent overfitting.

The MONAI framework allowed us to achieve good results by providing an extensive set of 3D models and transformations. In particular, we were able to omit the tedious task of creating a 3D implementation from scratch and instead utilized the tools provided by MONAI. According to Muhammad et al., the framework offers a straightforward method to import and modify existing networks for segmentation tasks, which was confirmed by our experiment [3]. Thus, we were able to employ the existing MONAI implementations of 3D UNet models and transformations for our task.

CONCLUSION

We present a volumetric 3D U-Net architecture for liver and tumor segmentation in CT scans. Unlike previous approaches based on 2D slices, our method exploits information from entire 3D volumes in order to perform accurate delineations. Preliminary experiments demonstrate that our approach achieves better tumor segmentations with improved boundaries. Future work will focus on expanding the model to include more complex attention mechanisms and multi-scale feature fusion, as well as testing on additional datasets for robustness. We also intend to make our code available to the research community, and provide trained networks for practical applications where fast and reliable segmentations are required.

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